

Random Matrices

June 12, 2026

—
Definition 0.1. Consider a matrix of random entries:

$$\begin{pmatrix} * & \cdots & * \\ \vdots & \ddots & \vdots \\ * & \cdots & * \end{pmatrix}$$

These entries are random variables. Let $X_1, \dots, X_n$ be vectors in $\mathbb{R}^d$, where each $X_i = (X_{i1}, \dots, X_{id})$ consists of random variables.

Definition 0.2. For a matrix X with entries X_{ij} :

1. The **variance** of the i -th component is given by:

$$\frac{1}{n} \sum_{i=1}^n X_{i,i}^2$$

2. The **covariance** between components i_1 and i_2 is given by:

$$\frac{1}{n} \sum_{i=1}^n X_{i,i_1} X_{i,i_2}$$

Let X be the data matrix:

$$X = \begin{pmatrix} X_{11} & \cdots & X_{1d} \\ \vdots & \ddots & \vdots \\ X_{n1} & \cdots & X_{nd} \end{pmatrix}$$

The sample covariance matrix is defined as:

$$\frac{XX^\top}{n}$$

Definition 0.3. The **eigenvalues** of a matrix A are defined by the equation:

$$Av = \lambda v \quad \text{for } v \neq 0.$$

The trace of A is given by:

$$\text{tr}(A) = \sum_i A_{ii} = \sum_i \lambda_i,$$

where λ_i are the eigenvalues of A .

Remark 0.1. The set of eigenvalues of a random matrix is called the **empirical spectral distribution**. This is given by:

$$\frac{1}{n} \sum_{i=1}^n \delta_{\lambda_i},$$

where δ_{λ_i} is the Dirac delta function centered at λ_i .

—

The handwritten content in the image does not explicitly contain definitions, theorems, proofs, or other structured mathematical statements. Instead, it appears to be a set of notes on properties of a matrix A with entries that are probabilistic and symmetric.

$$\begin{pmatrix} A_{11} & A_{1j} & \cdots & \\ A_{j1} & A_{jj} & \cdots & \\ \vdots & \vdots & \ddots & \\ A_{m1} & A_{mj} & \cdots & A_{mn} \end{pmatrix}$$

where the entries A_{ij} are probabilistic with $\mathbb{P}(A_{ij} = 1) = \mathbb{P}(A_{ij} = -1) = \frac{1}{2}$ and $A_{ji} = A_{ij}$.

$$\text{tr}(A) = \sum_i A_{ii} = \sum_i \lambda_i$$

$$\text{tr}(A^2) = \sum_i A_{ii}^2 = \sum_i \lambda_i^2$$

$$\text{tr}(A^k) = \sum_i A_{ii}^k = \sum_i \lambda_i^k$$